

Part I

SUMMARY of Definite Integrals

Summary

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Definition

If function f is continuous on interval $[a, b]$ then

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x$$

Note 1: When $f(x) \geq 0$, on the interval $[a, b]$, the definite integral, $\int_a^b f(x) dx$, gives the **area** of the region between the graph of f and the interval $[a, b]$ on the x -axis.

Note 2: When $f(x) < 0$, on some interval in $[a, b]$, then the definite integral, $\int_a^b f(x) dx$, gives the **net area** of the region between the graph of f and the interval $[a, b]$ on the x -axis.

Properties of Definite Integrals

$$(a) \int_a^a f(x) dx = 0$$

$$(b) \int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx$$

$$(c) \int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$(d) \int_a^b k \cdot f(x) dx = k \cdot \int_a^b f(x) dx$$

$$(e) \int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx; \int_a^b (f(x) - g(x)) dx = \int_a^b f(x) dx - \int_a^b g(x) dx$$

Note 3: If the function f is **odd**, i.e, if $f(-x) = -f(x)$, for all x in $[-a, a]$, then

$$\int_{-a}^a f(x) dx = 0$$

Note 4: If the function f is **even**, i.e, if $f(-x) = f(x)$, for all x in $[-a, a]$, then

Summary

$$\int_{-a}^a f(x) \, dx = 2 \cdot \int_0^a f(x) \, dx$$

Part II

Recitation Questions

Problem 1

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Problem 1

Problem 1 Consider the following limit of Riemann sums of a function g on $[a, b]$:

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n (x_k^* + \cos(x_k^*)) \Delta x, [0, \pi].$$

Express the limit as a definite integral. Use geometry to evaluate the resulting definite integral.

Problem 2

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Problem 2

Problem 2 Let $f(x)$ and $g(x)$ be functions for which we only know the following:

$$\int_1^4 f(x) dx = 7 \quad \int_2^4 f(x) dx = 5 \quad \int_1^4 g(x) dx = 2$$

Compute the following integrals, if possible. If it is not possible, give examples explaining why not.

(a) $\int_1^4 (8f(x) - 7g(x)) dx$

(b) $\int_1^2 (-f(x)) dx$

(c) $\int_1^4 |f(x)| dx$

(d) $\int_1^4 (2 - x + f(x)) dx$

Problem 3

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Problem 3

Problem 3 *Evaluate the following sums:*

(a) $\sum_{k=1}^4 k^5$

(b) $\sum_{k=1}^{400} (5(k+1)^2 + 3)$

Problem 4

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Problem 4

Problem 4 Use geometry to evaluate the definite integral. Sketch the graph of the function and shade the relevant regions.

(a) $\int_1^3 (2x - 4) \, dx$

(b) $\int_1^3 |2x - 4| \, dx$

(c) $\int_0^1 (2x - 4) \, dx$

(d) $\int_{-1}^3 \sqrt{4 - (x - 1)^2} \, dx$

Problem 5

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Problem 5

Problem 5 (a) If f is an odd function, why is it true that $\int_{-a}^a f(x) dx = 0$?
Support your reasoning with a picture.

(b) If f is an even function, why is it true that $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$?
Support your reasoning with a picture.

Problem 6

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Problem 6

Problem 6 (a) Find the following definite integral:

$$\int_{-4}^4 \frac{x^2 \sin^3(x)}{\sqrt{x^4 + 1}} dx$$

(b) Suppose that f is an even function. Given that $\int_0^6 f(x) dx = 13$, find

$$\int_{-6}^6 (5f(x) + 14) dx.$$

Problem 7

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Problem 7

Problem 7 Evaluate the following integrals using symmetry arguments.

(a) $\int_{-\pi/4}^{\pi/4} \sin(t) \, dt$

(b) $\int_{-2}^2 (1 + x + 3x^7 - x^9) \, dx$

(c) $\int_{-\pi}^{\pi} x \cos(x) \, dx$